

Advanced Time Series Analysis and Forecasting

Seminar 2: Structural breaks and nonlinear models



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BIS**: Bank for International Settlements
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **COVID**: COronaVIrus Disease (2019 pandemic)
- **CPI**: Consumer Price Index
- **CRPS**: Continuous Ranked Probability Score
- **CSW**: Chu, Stinchcombe and White (monitoring boundary)
- **DF**: Dickey–Fuller (test)
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **ESTAR**: Exponential Smooth Transition AutoRegression
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GW**: Giacomini–White (test of conditional predictive ability)

Acronyms used in this material (2/2)

- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **ICSS**: Iterated Cumulative Sums of Squares
- **IT**: Inclán–Tiao (statistic of the ICSS algorithm)
- **KSS**: Kapetanios, Shin and Snell (unit-root test against ESTAR)
- **LM3**: Lagrange Multiplier linearity test, third-order expansion
- **LR**: Likelihood Ratio (test)
- **LSTAR**: Logistic Smooth Transition AutoRegression
- **LWZ**: Liu, Wu and Zidek (modified Schwarz criterion)
- **MSFE**: Mean Squared Forecast Error
- **OLS**: Ordinary Least Squares
- **PPP**: Purchasing Power Parity
- **RMSE**: Root Mean Squared Error
- **SCA**: Seasonally and Calendar Adjusted
- **SETAR**: Self-Exciting Threshold AutoRegression
- **SSR**: Sum of Squared Residuals
- **STAR**: Smooth Transition AutoRegression
- **TAR**: Threshold AutoRegression
- **US**: United States

Today's question and seminar structure

- The question of the seminar
  - **how do we date breaks and test for nonlinearity when the break date or the threshold is not known in advance?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the Chow test; the Wald process of a mean shift	Solved, Proposed	A1
A3, A4	dynamic programming by hand; BIC, LWZ and sequential tests	Solved, Proposed	A3
A5, A6	a SETAR skeleton and the threshold LR; ESTAR persistence	Solved, Proposed	A5
A7, A8	the monitoring boundary and κ_2; the optimal window	Solved, Proposed	A7
B1, B2	Romanian GDP volatility; Bai–Perron on the 3-month rate	Solved, Proposed	B1
B3–B5	SETAR for sunspots; ESTAR for the RON real exchange rate; BET variance breaks	Solved, Proposed	B1, B3
C1, C2	window choice for inflation forecasts; what is wrong in an AI answer?	Proposed	A8, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; its solution is discussed in the seminar
- **A**: calculations and derivations on paper; **B**: real data, solved in the notebook, each closing with an interpretation question; **C**: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
real GDP, Romania (SCA)	Eurostat (namq_10_gdp)	quarterly	1995–2026
3-month money-market rate, Romania	Eurostat (irt_st_m)	monthly	2005–2026
sunspot numbers	statsmodels data set	yearly	1700–2008
real effective exchange rate, Romania	BIS via FRED (RBROBIS)	monthly	1994–2026
BET	EODHD	daily	2000–2026
HICP inflation, Romania	Eurostat (prc_hicp_minr)	monthly	2000–2026

- In the notebook: `andrews_tests`, `bp_analysis`, `tar_estimate`, `hansen_test`, `lm3_test`, `estar_fit`, `icss`, `csw_boundary`; no account or key is needed
- BIS: Bank for International Settlements; SCA: seasonally and calendar adjusted

Prerequisites for Today (1/22): regression with a break

- Details: Chapter 2
- Linear model with coefficients that may change over time

$$y_t = x_t'\beta_t + u_t, \qquad t = 1, \ldots, T$$

  - x_t: k regressors (a constant, lags of y_t); β_t: the coefficients; u_t: the error
  - **one break** at $T_1 = [\pi T]$: $\beta_t = \beta_1$ for $t \leq T_1$ and β_2 after; $\pi \in (0, 1)$: the break fraction; $[\cdot]$: the integer part
- **SSR** (sum of squared residuals) of an OLS fit on a sample

$$\mathrm{SSR} = \sum_t (y_t - x_t'\hat{\beta})^2$$

  - a break model fits each segment separately; its SSR can only be smaller than the SSR of one pooled fit

Prerequisites for Today (2/22): the Chow test

- **Chow** Chow (1960): break at a **known** date T_1; S_0: SSR of the pooled fit, S_1, S_2: SSR of the two sub-samples

$$F = \frac{(S_0 - S_1 - S_2)/k}{(S_1 + S_2)/(T - 2k)} \sim F(k, T - 2k)$$

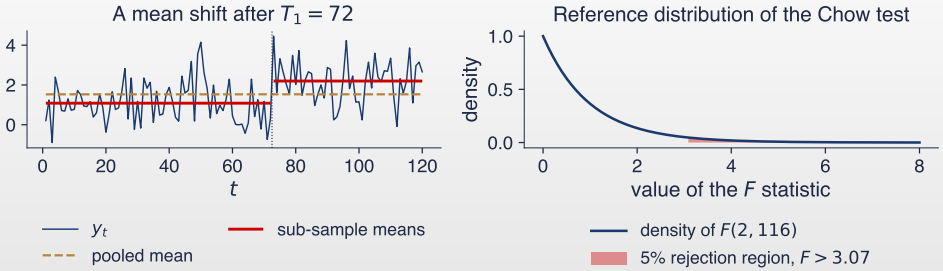
  - numerator: the gain in fit per extra coefficient; denominator: the residual variance of the break model
  - $F(d_1, d_2)$: the ratio of two independent χ^2 variables, each divided by its degrees of freedom; a large F is evidence of a break

- **Predictive** Chow test: $n_2 = T - T_1$ new observations, too few to estimate a second model

$$F = \frac{(S_0 - S_1)/n_2}{S_1/(T_1 - k)} \sim F(n_2, T_1 - k)$$

  - does the model estimated up to T_1 still fit the new observations?

Prerequisites for Today (3/22): a mean shift and the F distribution



- Left: a simulated mean shift; the two sub-sample means fit much better than the pooled mean
- Right: the Chow statistic is compared with the 95% quantile of $F(k, T - 2k)$, here $k = 2$, $T = 120$

Prerequisites for Today (4/22): one break at an unknown date

- **Wald statistic** of no break at fraction π (R: selects $\beta_{\mathbf{1}} - \beta_{\mathbf{2}}$; $\hat{V}$: the covariance of the estimates)

$$W_T(\pi) = (R\hat{\theta})'\left(R\hat{V}R'\right)^{-\mathbf{1}}(R\hat{\theta}), \qquad \hat{\theta} = (\hat{\beta}_{\mathbf{1}}', \hat{\beta}_{\mathbf{2}}')'$$

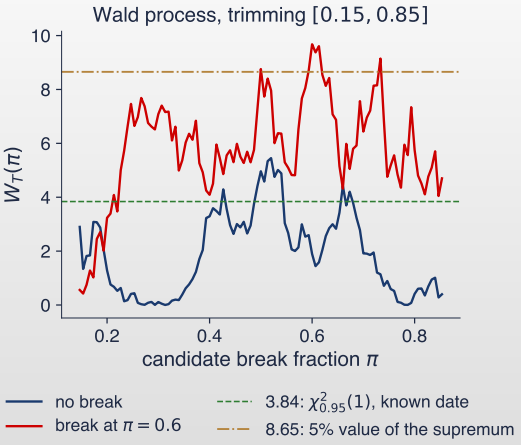
  - at a known date $W_T \sim \chi^{\mathbf{2}}(k)$; $W_T \approx kF$

- Unknown date Andrews (1993), Andrews and Ploberger (1994): combine $W_T(\pi)$ over $\Pi = [0.15, 0.85]$ (15% trimming at each end)

$$\sup W = \max_{\pi \in \Pi} W_T(\pi), \quad \exp W = \ln\left[\frac{\mathbf{1}}{|\Pi|}\sum_{\pi\in\Pi} e^{W_T(\pi)/\mathbf{2}}\right], \quad \text{ave}W = \frac{1}{|\Pi|}\sum_{\pi\in\Pi} W_T(\pi)$$

  - $|\Pi|$: the number of candidate dates; the estimated date is the arg max of $W_T(\pi)$
  - the search raises the critical value: 5% value of $\sup W$ (one coefficient): 8.65, against 3.84 at a known date

Prerequisites for Today (5/22): the Wald process



- $W_T(\pi)$ for two simulated series, $T = 150$
- Blue: no break
  - the process often crosses 3.84 somewhere: comparing its maximum with 3.84 would find false breaks
- Red: a mean shift at $\pi = 0.6$
  - the peak is near the true date and above the sup-Wald critical value
- Near $\pi = 0$ or 1 one mean rests on very few observations: hence the trimming

Prerequisites for Today (6/22): mean and variance breaks

- **Mean shift** with known variance σ^2: $y_t = \mu + \delta 1\{t > [\pi T]\} + u_t$

$$W_T(\pi) = T\pi(1-\pi)\frac{(\bar{y}_1 - \bar{y}_2)^2}{\sigma^2}$$

  - $\bar{y}_1$, $\bar{y}_2$: the means before and after $[\pi T]$; δ: the size of the shift
- **Variance break**, two steps McConnell and Perez-Quiros (2000): fit the model, then test the mean of a scale proxy

$$v_t = \sqrt{\pi/2}\,|\hat{e}_t|, \qquad \mathbb{E}v_t = \sigma \text{ if } e_t \sim N(0, \sigma^2)$$

  - $\hat{e}_t$: the residuals; here $\pi = 3.14\ldots$ (not the break fraction); a break in the mean of v_t is a break in the standard deviation
  - use HAC or heteroskedasticity-robust variances in the Wald statistics: v_t is autocorrelated when volatility clusters

Prerequisites for Today (7/22): several breaks: Bai–Perron

- m breaks $T_1 < \cdots < T_m$ split the sample into $m+1$ segments, each with its own β_j
  - **Bai–Perron** Bai and Perron (1998), Bai and Perron (2003): the dates minimise the total SSR over all partitions
  - each segment has at least $h = [\varepsilon T]$ observations (ε: trimming, e.g. 0.15); at most M breaks
- **Dynamic programming**: the best r-break partition of the first n observations

$$\mathrm{SSR}_r(n) = \min_j \left[\mathrm{SSR}_{r-1}(j) + \mathrm{SSR}(j+1, n)\right]$$

  - $\mathrm{SSR}(i, n)$: the SSR of one regression on observations $i, \ldots, n$; j: the position of the last break
  - reuses the solutions for $r-1$ breaks: about T^2 regressions instead of all $\binom{T}{m}$ partitions

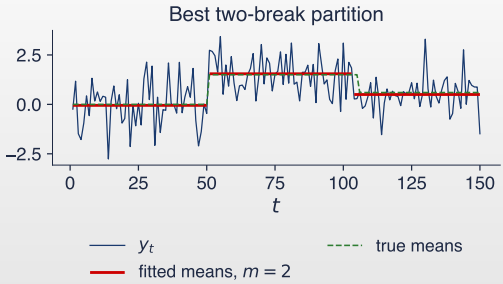
Prerequisites for Today (8/22): how many breaks

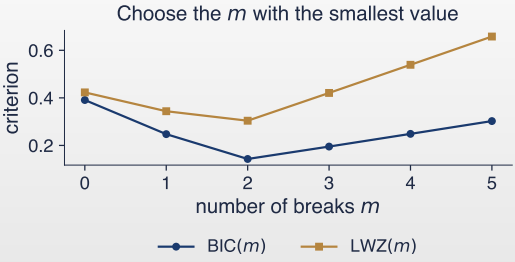
- **Tests** with robust variances
  - $\sup F(k)$: 0 breaks against k breaks; UDmax $= \max_{k \le M} \sup F(k)$: 0 against an unknown number
  - sequential $F(\ell+1 \mid \ell)$: with ℓ breaks fixed, does one more break, inside some segment, reduce the SSR significantly? Add breaks while it rejects
- **Information criteria** with $p^* = 2m+1$ parameters for m mean shifts ($m+1$ means, m dates)

$$\mathrm{BIC}(m) = \ln \frac{\mathrm{SSR}_m}{T} + \frac{p^* \ln T}{T}, \qquad \mathrm{LWZ}(m) = \ln \frac{\mathrm{SSR}_m}{T - p^*} + \frac{0.299\, p^* (\ln T)^{\mathbf{2.1}}}{T}$$

  - SSR_m: the SSR of the best m-break partition; the first term rewards fit, the second penalises parameters
  - choose the m with the smallest value; LWZ Liu, Wu and Zidek (1997) penalises extra breaks more than BIC

Prerequisites for Today (9/22): a two-break example





- Left: three simulated regimes and the best two-break partition found by dynamic programming
- Right: both criteria fall sharply from $m = 0$ to the true number of breaks, then flatten or rise

Prerequisites for Today (10/22): a confidence interval for the date

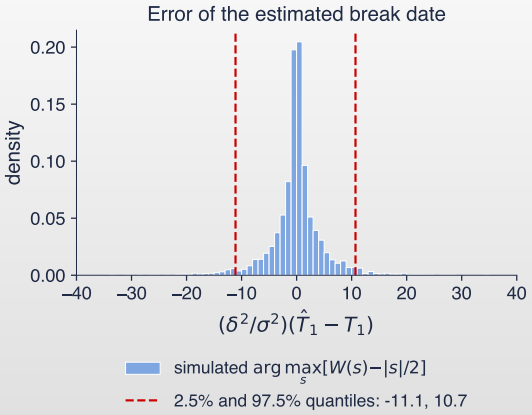
- **Bai** Bai (1997): a mean shift of size δ, error variance σ^2

$$\frac{\delta^2}{\sigma^2}\left(\hat{T}_1 - T_1\right) \xrightarrow{d} \arg\max_s \left[W(s) - \frac{|s|}{2}\right]$$

  - $\hat{T}_1$: the estimated date; W: a two-sided Brownian motion ($W(0) = 0$, independent Brownian motions for $s > 0$ and $s < 0$)
  - Brownian motion: independent Normal increments, $W(s) - W(r) \sim N(0, |s - r|)$

- Reading
  - with equal variances the 2.5% and 97.5% quantiles of the limit are about ∓ 11
  - 95% interval: $\hat{T}_1 \pm 11\,\sigma^2/\delta^2$ periods; small shifts are dated very imprecisely

Prerequisites for Today (11/22): the distribution of the date error



- 4000 simulated draws of $\arg\max_s[W(s) - |s|/2]$
- Peaked at 0, but with long tails: the date is usually close, sometimes far
- Example
  - $\delta = \sigma$: the interval is about ± 11 periods
  - $\delta = \sigma/2$: about ± 44 periods

Prerequisites for Today (12/22): monitoring for a break

- **Recursive residuals**: standardised one-step forecast errors of the model estimated up to $t-1$

$$w_t = \frac{y_t - x_t' \hat{\beta}_{t-1}}{\sqrt{1 + x_t'(X_{t-1}'X_{t-1})^{-1}x_t}}$$

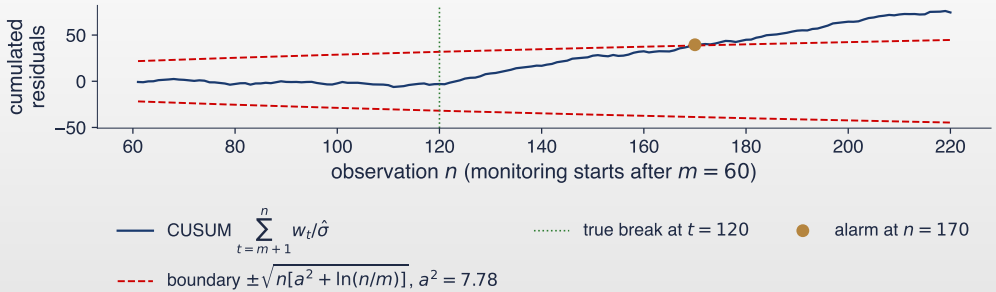
  - $\hat{\beta}_{t-1}$, X_{t-1}: the OLS estimate and the regressor matrix up to $t-1$; without a break w_t are i.i.d. $N(0, \sigma^2)$ under Normal errors

- **Monitoring** Chu, Stinchcombe and White (1996): after m historical observations, alarm at the first n with

$$\left| \sum_{t=m+1}^{n} w_t / \hat{\sigma} \right| > \sqrt{n\left[a^2 + \ln(n/m)\right]}$$

  - $\hat{\sigma}$: estimated on the historical sample; a: a constant that sets the size
  - probability of a false alarm over an infinite horizon: $2[1 - \Phi(a) + a\varphi(a)]$ Robbins and Siegmund (1970)

Prerequisites for Today (13/22): an example of a monitoring boundary



- A constant mean is estimated on $m = 60$ observations; a shift of 1σ occurs at $t = 120$
- The cumulated residuals drift away after the break and cross the boundary tens of periods later: monitoring has a detection delay

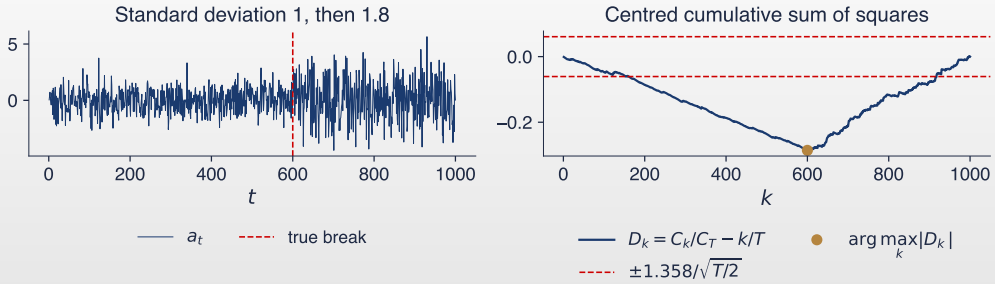
Prerequisites for Today (14/22): variance breaks: Inclán–Tiao and ICSS

- a_t: demeaned returns; centred cumulative sum of squares Inclán and Tiao (1994)

$$C_k = \sum_{t=1}^{k} a_t^2, \qquad D_k = \frac{C_k}{C_T} - \frac{k}{T}, \qquad \mathrm{IT} = \sqrt{T/2}\, \max_k |D_k|$$

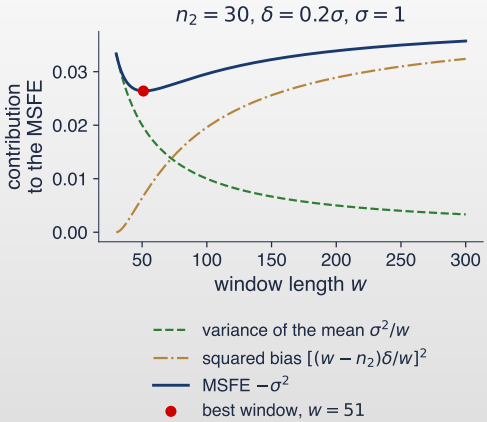
  - constant variance: C_k grows linearly and D_k stays near 0; 5% critical value of IT: 1.358
  - the factor $\sqrt{T/2}$ assumes $\mathrm{Var}(a_t^2) = 2\sigma^4$, true for i.i.d. Normal data
- Fat tails and volatility clustering
  - **kurtosis** $\kappa = \mathbb{E}a_t^4/\sigma^4$ ($\kappa = 3$ for the Normal); with i.i.d. data IT is too large by a factor $\sqrt{(\kappa-1)/2}$
  - κ_2 Sansó, Aragó and Carrion (2004): replaces $2\sigma^4$ by a long-run variance of a_t^2
- **ICSS** (iterated cumulative sums of squares): find the largest $|D_k|$
  - if significant, split the sample there and repeat in each piece until no piece rejects

Prerequisites for Today (15/22): a simulated variance break



- Left: the standard deviation rises from 1 to 1.8 at $t = 600$
- Right: D_k drifts down while the calm part accumulates too little, and turns at the break

Prerequisites for Today (16/22): the estimation window



- Mean shift δ that happened n_2 periods ago; forecast $\bar{y}_w$, the mean of the last $w \geq n_2$ observations Pesaran and Timmermann (2007)

$$\mathrm{MSFE}(w) = \sigma^2\left(1 + \frac{1}{w}\right) + \left[\frac{(w - n_2)\,\delta}{w}\right]^2$$

  - MSFE: the mean squared forecast error; σ^2: the error variance
  - longer window: less variance, more bias from pre-break data
- Schemes
  - expanding window; rolling window of fixed length
  - AveW Pesaran and Pick (2011): average the forecasts of all windows from a minimum length to all data

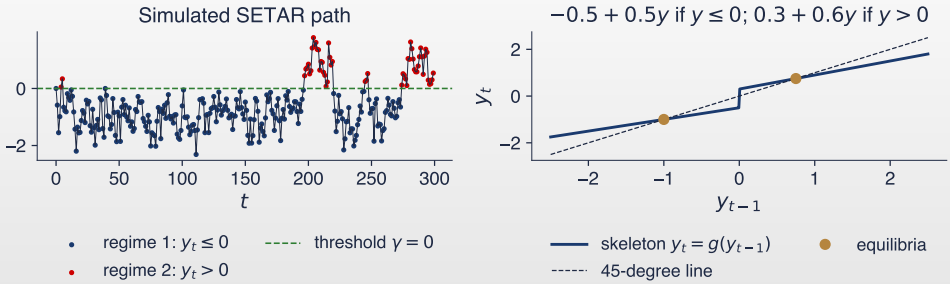
Prerequisites for Today (17/22): the SETAR model

- **SETAR** (self-exciting threshold AR) Tong and Lim (1980): the AR coefficients depend on the regime of y_{t-d}

$$y_t = \begin{cases} \phi_{\mathbf{1},\mathbf{0}} + \phi_{\mathbf{1}}' \mathbf{y}_{t-\mathbf{1}} + \varepsilon_t, & y_{t-d} \leq \gamma \\ \phi_{\mathbf{2},\mathbf{0}} + \phi_{\mathbf{2}}' \mathbf{y}_{t-\mathbf{1}} + \varepsilon_t, & y_{t-d} > \gamma \end{cases}$$

  - $\mathbf{y}_{t-\mathbf{1}} = (y_{t-\mathbf{1}}, \ldots, y_{t-p})'$; $\phi_{j,\mathbf{0}}$, ϕ_j: intercept and AR coefficients of regime j
  - γ: the threshold; d: the delay lag; SETAR$(2; p_{\mathbf{1}}, p_{\mathbf{2}})$: two regimes with AR orders $p_{\mathbf{1}}$, $p_{\mathbf{2}}$
- Dynamics
  - **skeleton**: the map $y_t = g(y_{t-\mathbf{1}}, \ldots)$ with $\varepsilon_t = 0$; it shows equilibria and cycles
  - regime mean $\phi_{j,\mathbf{0}}/(1 - \phi_j)$ in a SETAR$(2; 1, 1)$; it is an equilibrium only if it lies inside regime j
  - SETAR$(2; 1, 1)$ with $d = 1$ is ergodic if and only if $\phi_{\mathbf{1}} < 1$, $\phi_{\mathbf{2}} < 1$ and $\phi_{\mathbf{1}}\phi_{\mathbf{2}} < 1$

Prerequisites for Today (18/22): a SETAR path and its skeleton



- Left: the path switches between two levels; colours mark the regime of y_t
- Right: both regime means lie inside their regimes, so the skeleton has two stable equilibria, -1 and 0.75; the noise moves the path between them

Prerequisites for Today (19/22): inference on the threshold

- **Linearity test**: H_0: one regime; γ is not identified under H_0 (the Davies problem)
  - statistic $\sup_\gamma W(\gamma)$ over a grid of thresholds (15% trimming); non-standard distribution
  - p-value by the **fixed-regressor bootstrap** Hansen (1996): keep x_t, draw $y_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0,1)$, recompute $\sup_\gamma W^*$; p-value: the share of $\sup W^* > \sup W$
- **Confidence set for** γ Hansen (2000): invert the likelihood ratio

$$\mathrm{LR}(\gamma) = T\,\frac{\mathrm{SSR}(\gamma) - \mathrm{SSR}(\hat{\gamma})}{\mathrm{SSR}(\hat{\gamma})}, \qquad \{\gamma : \mathrm{LR}(\gamma) \le c(\alpha)\}, \quad c(\alpha) = -2\ln\left(1 - \sqrt{1-\alpha}\right)$$

  - $\mathrm{SSR}(\gamma)$: the SSR of the SETAR with threshold γ; $\hat{\gamma}$ minimises it

Prerequisites for Today (20/22): smooth transition models

- **STAR** Teräsvirta (1994): two AR regimes mixed by a weight $G(s_t; \gamma, c) \in [0, 1]$

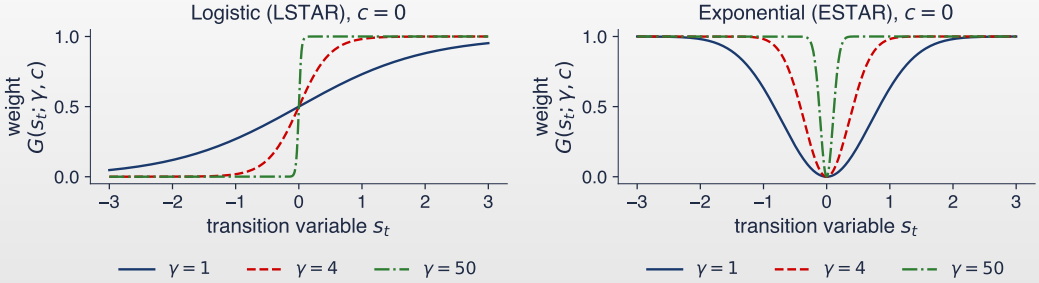
$$y_t = \phi_{\mathbf{1}}' w_t + \phi_{\mathbf{2}}' w_t \, G(s_t; \gamma, c) + \varepsilon_t, \qquad w_t = (1, y_{t-\mathbf{1}}, \ldots, y_{t-p})'$$

  - s_t: the transition variable (often y_{t-d}); c: the location; $\gamma > 0$: the speed of the transition
  - LSTAR: $G = [1 + e^{-\gamma(s_t - c)}]^{-\mathbf{1}}$ (low and high regimes); ESTAR: $G = 1 - e^{-\gamma(s_t - c)^{\mathbf{2}}}$ (inner and outer regimes)
- **LM3 linearity test**: a Taylor expansion of G around $\gamma = 0$ gives the auxiliary regression

$$y_t = \beta_{\mathbf{0}}' w_t + \beta_{\mathbf{1}}' w_t s_t + \beta_{\mathbf{2}}' w_t s_t^{\mathbf{2}} + \beta_{\mathbf{3}}' w_t s_t^{\mathbf{3}} + e_t$$

  - LM3: $\beta_{\mathbf{1}} = \beta_{\mathbf{2}} = \beta_{\mathbf{3}} = 0$; sequence $H_{\mathbf{04}} : \beta_{\mathbf{3}} = 0$, $H_{\mathbf{03}} : \beta_{\mathbf{2}} = 0 \mid \beta_{\mathbf{3}} = 0$, $H_{\mathbf{02}} : \beta_{\mathbf{1}} = 0 \mid \beta_{\mathbf{2}} = \beta_{\mathbf{3}} = 0$
  - the smallest p-value for $H_{\mathbf{03}}$ points to ESTAR, otherwise to LSTAR

Prerequisites for Today (21/22): logistic and exponential transitions



- Left: LSTAR moves from regime 1 ($G = 0$) to regime 2 ($G = 1$) as s_t crosses c; γ sets how abrupt the move is
- Right: ESTAR treats large deviations of either sign alike: regime 1 near c, regime 2 far from c

Prerequisites for Today (22/22): ESTAR mean reversion and the KSS test

- Real exchange rate $q_t = s_t + p_t^* - p_t$ (logs: nominal rate, foreign and domestic prices); PPP: q_t reverts to an equilibrium μ
  - PPP: purchasing power parity; transaction costs make small deviations persist
- ESTAR of Taylor, Peel and Sarno (2001)

$$q_t - \mu = (q_{t-1} - \mu)\, e^{-\theta^2 (q_{t-1} - \mu)^2} + \varepsilon_t$$

  - local AR coefficient $\rho(x) = e^{-\theta^2 x^2}$ at deviation x: close to 1 near μ, small far away
  - **half-life** of a deviation with AR coefficient ρ: $\ln 0.5 / \ln \rho$ periods (half of the gap closes)
- Unit-root tests
  - Dickey–Fuller: t of δ in $\Delta q_t = \delta q_{t-1} + e_t$; $\delta = 0$ is a unit root (a random walk)
  - KSS Kapetanios, Shin and Snell (2003): the same with q_{t-1}^3 (q demeaned); reject the unit root in the left tail of its own simulated distribution

A1: the Chow test by hand [Solved]

Task

- A regression with $k = 2$ coefficients on $T = 80$ quarters; break candidate after observation 40. SSR: pooled 120, first half 45, second half 50.
- 1. Compute the Chow statistic and its 5% critical value; conclude.
- 2. Six new quarters arrive; the SSR of the first 40 plus the six new ones is 52. Compute the predictive Chow statistic.
- 3. The date 40 was chosen after looking at a plot. Which critical value should replace the F one?
- Report: two statistics, two critical values, one sentence.

Solution

- 1. $F = \dfrac{(120 - 95)/2}{95/76} = 10.00$; $F_{\mathbf{0.95}}(2, 76) = 3.12$, $p < 0.001$: the coefficients change
- 2. $F = \dfrac{(52 - 45)/6}{45/38} = 0.985$, $F_{\mathbf{0.95}}(6, 38) = 2.35$, p 0.449: the model still forecasts the new quarters
- 3. the sup-Wald value for $p = 2$ restrictions (about 11.7 at 5% with 15% trimming); the Wald form is $2F = 20$: still a rejection

Back Exercise list

A2: the Wald process of a mean shift [Proposed]

Solution discussed in class

Task

- $y_t = \mu + \delta 1\{t > [\pi T]\} + u_t$, u_t i.i.d. $N(0, \sigma^2)$, $\sigma^2 = 1$ known, $T = 100$. Model: A1.
- 1. Show that the Wald statistic for $\delta = 0$ at fraction π is $W_T(\pi) = T\pi(1-\pi)(\bar{y}_1 - \bar{y}_2)^2/\sigma^2$.
- 2. The largest Wald statistic is found at $\pi = 0.3$ with $\bar{y}_1 - \bar{y}_2 = 0.5$. Compute it and compare with $\chi^2_{0.95}(1)$ and with the sup-Wald value.
- 3. Explain why the search range must exclude π close to 0 and 1.
- Report: one derivation, one statistic, one decision, one sentence.

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A3: dynamic programming on eight observations [Solved]

Task

- $y = (1, 1, 2, 1, 5, 6, 5, 6)$, mean-shift model, minimum segment length $h = 2$.
- 1. Compute $\mathrm{SSR}(i, j)$ for the segments needed by the one-break problem and find the best single break.
- 2. Use the recursion to find the best two-break partition.
- 3. How much does the second break reduce the SSR?
- Report: the SSR table, two partitions, one comparison.

Solution

- 1. no break: SSR 37.875; candidates: $[1, 2] + [3, 8]$: $0.000 + 22.833$; $[1, 3] + [4, 8]$: $0.667 + 17.200$; $[1, 4] + [5, 8]$: $0.750 + 1.000 = 1.75$; $[1, 5] + [6, 8]$: $12.000 + 0.667$; $[1, 6] + [7, 8]$: $25.333 + 0.500$
- best break after observation 4 (means 1.25 and 5.5)
- 2. $\mathrm{SSR}_{\mathbf{2}}(8) = \min_j [\mathrm{SSR}_{\mathbf{1}}(j) + \mathrm{SSR}(j + 1, 8)]$; the minimum is $[1, 2], [3, 4], [5, 8]$: $0 + 0.5 + 1 = 1.50$
- 3. from 1.75 to 1.50: a small gain that BIC would not buy; the first break explains almost everything

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A4: BIC, LWZ and the sequential tests [Proposed]

Solution discussed in class

Task

- The US ex-post real interest rate, quarterly, 1961Q1–1986Q3 (the data of Bai and Perron (2003)): $T = 103$, SSR for $m = 0, 1, 2, 3$ mean shifts: 1214.9, 645.0, 456.0, 445.2. Model: A3.
- 1. Compute BIC(m) and LWZ(m) for $m = 0, \ldots, 3$ with $p^* = 2m + 1$ and choose m by each criterion.
- 2. The sequential statistics are $F(2|1) = 36.86$, $F(3|2) = 16.09$, $F(4|3) = 0.04$ with 5% values 10.54, 11.56, 11.96. How many breaks does the sequential procedure select?
- 3. Explain why the two approaches can disagree.
- Report: a table of eight values, two choices, one sentence.

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A5: a SETAR skeleton and the threshold LR [Solved]

Task

- SETAR(2; 1, 1) with $d = 1$, threshold 0: $y_t = 0.4 + 0.5y_{t-1}$ if $y_{t-1} \le 0$; $y_t = -0.6 + 0.8y_{t-1}$ if $y_{t-1} > 0$ (plus noise).
- 1. Is the process ergodic? Compute the regime means and say whether they lie in their own regimes (Prerequisites (17/22)).
- 2. Iterate the skeleton from $y_0 = 1$ for five steps and find its limit.
- 3. Derive the 95% and 90% critical values of the threshold LR from $P(\xi \le x) = (1 - e^{-x/2})^2$.
- Report: one condition, five numbers, a limit, two critical values.

Solution

- 1. $\phi_1 = 0.5 < 1$, $\phi_2 = 0.8 < 1$, $\phi_1\phi_2 = 0.4 < 1$: ergodic; regime means 0.8 (regime 1 needs $y \le 0$) and -3.0 (regime 2 needs $y > 0$): both are *virtual*
- 2. 0.200, -0.440, 0.180, -0.456, 0.172: a two-cycle $a = (0.4 - 0.5 \cdot 0.6)/(1 - 0.4) = 0.167$, $b = -0.6 + 0.8a = -0.467$
- 3. $(1 - e^{-x/2})^2 = 1 - \alpha \Rightarrow x = -2\ln(1 - \sqrt{1 - \alpha})$: 7.35 (95%) and 5.94 (90%)
- a nonlinear model can oscillate forever without noise; no linear AR(1) can

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A6: local persistence in an ESTAR [Proposed]

Solution discussed in class

Task

- The dollar–sterling ESTAR of Taylor, Peel and Sarno (2001): $\theta^2 = 0.452$; a linear AR(1) fitted to the same data has a half-life of about 45 months. Model: A5.
- 1. Compute the local AR coefficient $\rho(x)$ of Prerequisites (22/22) for deviations $x = 0.05, 0.20, 0.40$ (5%, 20%, 40%).
- 2. Compute the implied "local" half-lives and the AR(1) coefficient that gives 45 months.
- 3. Explain why a Dickey–Fuller test has low power when the data follow this ESTAR.
- Report: three coefficients, four half-lives, one sentence.

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A7: the monitoring boundary and the κ_2 correction [Solved]

Task

- Monitoring with $m = 60$ historical months; ICSS on $T = 500$ returns with $C_{250}/C_{500} = 0.70$.
- 1. Compute the asymptotic false-alarm probability of Prerequisites (12/22) for $a^2 = 7.78$ and $a^2 = 6.25$.
- 2. Compute the 5% boundary (same slide) at $n = 90$ and $n = 120$.
- 3. Compute the IT statistic at $k = 250$; if the returns are i.i.d. with kurtosis 6, compute the corrected statistic.
- Report: two sizes, two boundaries, two statistics, one decision.

Solution

- 1. $a = 2.789$: 5.08%; $a = 2.5$: 10.01%
- 2. $\sqrt{90(7.78 + \ln 1.5)} = 27.14$; $\sqrt{120(7.78 + \ln 2)} = 31.89$: the boundary widens like $\sqrt{n \ln n}$
- 3. $D_{250} = 0.70 - 0.50 = 0.20$, $\mathrm{IT} = \sqrt{250} \cdot 0.20 = 3.16$; factor $\sqrt{(6-1)/2} = 1.58$, corrected $2.00 > 1.358$: still a break, but much weaker evidence

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A8: the optimal estimation window [Proposed]

Solution discussed in class

Task

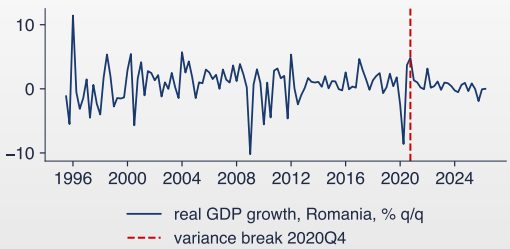
- A mean shift $\delta = 0.5\sigma$, $\sigma = 1$, happened $n_2 = 20$ periods ago; we forecast with the mean of the last $w \in [20, 200]$ observations. Model: A7.
- 1. Derive $\mathrm{MSFE}(w) = \sigma^2(1 + 1/w) + [(w - n_2)\delta/w]^2$.
- 2. Find the optimal w and compare its MSFE with $w = 20$ and $w = 200$.
- 3. How does the optimal window change with δ?
- Report: one derivation, three MSFEs, one sentence.

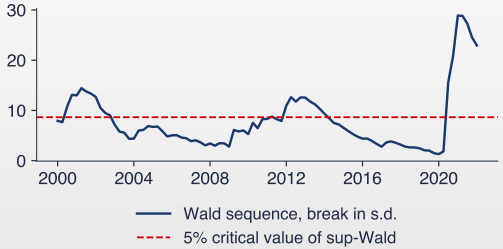
B1: a Great Moderation in Romania? [Solved]

- **Question**: has the volatility of Romanian GDP growth fallen, and when?
- **Inputs**: real GDP growth q/q (%, Eurostat, SCA), 1995Q3–2026Q2 ($T = 124$); the two-step method of McConnell and Perez-Quiros (2000)
- **Tasks**
  1. Fit an AR(1) and test for a break in its coefficients with sup-Wald (robust variance).
  2. Test for a break in the mean of the scale proxy v_t of Prerequisites (6/22) with sup-, exp- and ave-Wald (HAC) and date it.
  3. Repeat the variance test without the four quarters of 2020.
  4. Interpretation: is the estimated date a change of regime or the effect of a few quarters?
- **Report**: two sup-Wald statistics with p-values, a date, two standard deviations, one sentence
- Notebook: section B1

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B1: solution [Solved]





- AR(1): $\hat{\phi} = -0.05$; break in the coefficients: sup-Wald 3.32 (p 0.838): none
- Variance: sup 28.9 ($p < 0.001$), exp 10.9, ave 7.9 (5% value of sup: 8.65); date 2020Q4; s.d. 3.00 before, 0.96 after
- Without 2020: sup 34.0 ($p < 0.001$), date 2012Q4
- Interpretation: volatility did fall, but the date depends on two COVID quarters; without them the break moves to 2012, after the post-crisis adjustment: report both and let a second break or a robust scale decide

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B2: regimes of the Romanian 3-month rate [Proposed]

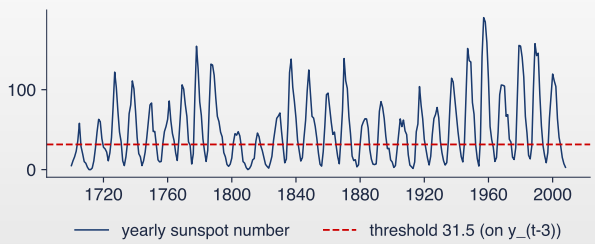
- **Question**: how many mean regimes does the Romanian money-market rate have since 2005, and do the tests agree?
- **Inputs**: 3-month rate, monthly (Eurostat irt_st_m), January 2005–August 2026 ($T = 260$); mean shifts, $\varepsilon = 0.15$, $M = 5$; model: B1 and A4
- **Tasks**
  1. Estimate the optimal partitions for $m = 1, \ldots, 5$ by dynamic programming.
  2. Report $\sup F(1)$, UDmax and the sequential statistics (robust variance) and the number of breaks they select.
  3. Report the numbers of breaks chosen by BIC and LWZ, the dates, the regime means and the 95% intervals.
  4. Interpretation: which partition would you report to the BNR, and why?
- **Report**: three test results, three choices of m, dates with intervals, one sentence
- Notebook: section B2

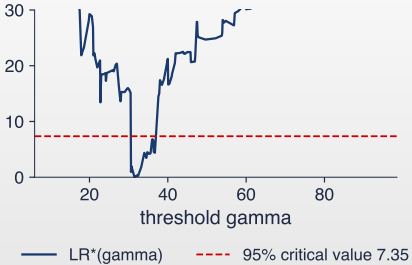
B3: a SETAR for the sunspots [Solved]

- **Question**: is the sunspot cycle better described by a two-regime SETAR than by a linear AR(3)?
- **Inputs**: yearly sunspot numbers 1700–2008 (statsmodels); SETAR(2; 3, 3), $d = 1, \ldots, 5$; trimming 15%; a series also studied by Tong and Lim (1980); model: A5
- **Tasks**
  1. Choose the delay d by the smallest SSR over the threshold grid.
  2. Test linearity with the heteroskedasticity-robust sup-Wald and the fixed-regressor bootstrap.
  3. Estimate the threshold and its 95% LR interval; compare the AR(1) and AR(2) coefficients of the two regimes.
  4. Interpretation: what does the threshold separate in the solar cycle?
- **Report**: five SSRs, one bootstrap p-value, a threshold with its interval, one sentence
- Notebook: section B3

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B3: solution [Solved]





- SSR for $d = 1, \ldots, 5$: 74 620, 58 448, 57 021, 72 363, 74 201: $\hat{d} = 3$; the SETAR reduces the SSR of the AR(3) by 31%
- sup-Wald 94.1, bootstrap $p < 0.001$ (500 replications); $\hat{\gamma} = 31.5$, 95% set [30.6; 36.7]; 123 and 181 years
- Lower regime: AR(1) 1.91, AR(2) -1.78 (strongly oscillatory); upper regime: 0.86, 0.11
- Interpretation: the threshold separates the quiet phase (low activity three years earlier), where the next rise is fast and regular, from the active phase, where the decline is slower: the asymmetry of the solar cycle

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B4: nonlinear mean reversion of the RON real exchange rate [Proposed]

- **Question**: does the Romanian real effective exchange rate revert to a mean nonlinearly, and how fast?
- **Inputs**: log real effective exchange rate (BIS, CPI-based, via FRED RBROBIS), January 1994–July 2026 ($T = 391$); model: B3 and A6
- **Tasks**
  1. Run the LM3 linearity test with $s_t = q_{t-\mathbf{1}}$ and the sequence $H_{\mathbf{04}}$, $H_{\mathbf{03}}$, $H_{\mathbf{02}}$.
  2. Compute the KSS statistic and its simulated 5% value.
  3. Estimate the ESTAR of Taylor, Peel and Sarno (2001) and the half-lives for shocks of 1%, 10% and 40%; compare with a linear AR(1).
  4. Interpretation: is ESTAR the right family for this series?
- **Report**: four p-values, one KSS test, two parameters, four half-lives, one sentence
- Notebook: section B4

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B5: variance regimes of the BET index [Proposed]

- **Question**: how many breaks in the unconditional variance do BET returns have, once fat tails and volatility clustering are accounted for?
- **Inputs**: daily log returns of the BET index (EODHD), 2000–2026 ($T = 6\,670$); model: A7
- **Tasks**
  1. Run ICSS with the Inclán–Tiao statistic and count the breaks.
  2. Run ICSS with the κ_2 statistic and report the dates and the standard deviation of each regime.
  3. Compare the two whole-sample statistics with the 5% value 1.358.
  4. Interpretation: which set of breaks would you give to a risk manager, and why?
- **Report**: two counts, the κ_2 dates, the regime standard deviations, one sentence
- Notebook: section B5

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C1: the window for twelve-month inflation forecasts [Proposed]

- **Question**: does a rolling or a window-averaged estimation beat the expanding window for Romanian inflation twelve months ahead, after correcting for three comparisons?
- **Inputs**: HICP inflation, y/y; direct AR(2) forecasts, targets January 2015–August 2026 ($P = 140$); windows: expanding, rolling 60 and 120 months, AveW (36 months to all); model: A8
- **Tasks**
  1. Write the pre-registration (target, horizon, loss, windows, test, correction) in five lines before computing anything.
  2. Compute the RMSE of the four schemes, using at each origin only targets already observed.
  3. Test each scheme against the expanding window with DM–HLN ($h = 12$) and apply the Holm correction (Holm (1979)).
  4. Interpretation: why can the ranking of the windows at twelve months differ from their ranking one month ahead?
- **Report**: a five-line plan, four RMSEs, three HLN statistics with raw and Holm p-values, a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise breaks and threshold models. The answer:
- (a) The 5% critical value of the Andrews sup-Wald test with one restriction and 15% trimming is 3.84.
- (b) The Bai-Perron break fractions converge at rate sqrt(T), like the regression coefficients.
- (c) A SETAR in which one regime is explosive on its own can never be stationary.
- (d) The 95% confidence set for a threshold collects the values with LR(gamma) <= 3.84.
- (e) A 5% test repeated at every new observation keeps the overall false-alarm rate at 5%.
- (f) With fat tails the Inclan-Tiao statistic finds too many variance breaks; kappa-2 corrects it with the long-run variance of the squared returns.
- (g) In an LSTAR, letting gamma tend to 0 gives a threshold model.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

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Key takeaways

- The Chow test is valid only for a date fixed in advance; searched dates need sup-type critical values
- Dynamic programming gives all optimal partitions; tests and criteria may disagree on their number
- Threshold models: bootstrap the linearity test, invert LR with 7.35, look at the skeleton
- ESTAR persistence depends on the size of the deviation; trends and breaks can mimic it
- An AI answer is a draft: check the critical values, the rates of convergence and the limits of the transition function

After the seminar

- Lecture 2 develops each topic of today: sup-Wald and its limit, Bai–Perron, monitoring, variance breaks, unit roots with breaks, windows, TAR, STAR and nonlinear forecasting
- C1 can grow into a team project: forecasting Romanian inflation across regimes
- Reading: Andrews (1993); Bai and Perron (2003); Hansen (1997); Taylor, Peel and Sarno (2001)
- **The seminar is for practice and is not graded**

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